

Compak Filter Operation & Maintenance Manual

SAFETY &/or HAZARD INFORMATION

This bulletin includes important safety information that should be read by owners, managers, service personnel, and anyone in charge of the pool or pool area. Also, we suggest a copy be posted for quick reference.

1. Only personnel trained and familiar with the proper use of pool chemicals should handle acid, liquid chlorine or chlorine compounds. Chemicals should never be used when swimmers are in the pool. Acid and liquid chlorine should always be stored, carried, or handled in plastic containers.
2. If grating is a part of the perimeter system, it should be kept firmly always clamped down and in good repair. When a section of grating becomes loose or damaged that particular area must be immediately covered and a replacement of grating ordered. Under no circumstances should swimmers be allowed to use any portion of the perimeter that contains loose or damaged grating. Perimeter grating is not intended for foot traffic. Swimmers should be advised not to walk, stand, or jump on perimeter grates.
3. Ladders and grab rails are intended for the use of one swimmer at a time; they are not designed for handstands or other gymnastic stunts and they should not be used for this purpose. Ladder treads should be inspected regularly. If a tread becomes loose or damaged, the ladder should be taken out of service until repairs are made.
4. Lifeguard Chairs are intended for the individual use of trained "on duty" lifeguards, one (1) guard per chair. Lifeguard chairs are not to be used by swimmers, spectators, or by more than one (1) person at a time. There should be no diving from portable lifeguard chairs. Umbrellas should be closed or removed from portable lifeguard chairs during windy conditions. All frame connections are to be checked for tightness. The seat is bolted to the frame assembly. It is important to advise all users to periodically check to determine that the studs are firmly fastened to the seat and the nuts are tight. If they become loose or detached, it could result in serious injury. On outdoor installations or usage, it is suggested that the seat be removed and stored inside during the winter.
5. Starting Platforms should only be used by trained competitive swimmers or under the direct supervision of an instructor. Swimmers should execute shallow racing dives only. Impact with the pool bottom can cause severe injury. Starting platforms have warning labels and inform the purchaser of the need to remove the platforms during non usage. If your starting platforms do not have warning labels, please contact the manufacturer immediately.
6. Bulkheads are designed and built for strength and safety. Any grating should be kept fully secured to avoid injury. **NO swimming under bulkhead.** **Never use bulkhead as a support or staging for equipment.** The bulkhead includes a compressor; please refer to owner's manual provided with the unit.

For questions concerning the usage of our equipment, please contact Paddock Pool Equipment Co., customer service.

Notes and Tips - From the Engineering Department

General notes:

1. Filter aid(s) such as alum are generally not required. Consult manufacture before use.
2. Clean and repack the recirculation pump and/or check the seal and overhaul the motor at least once a year.
3. Establish definite periods for using the vacuum cleaner and skimmer in removing leaves and other foreign matter from the pool.
 - 3.1. Do not allow nails, pins or other metal articles to remain in the pool for any length of time, as a rust stain will mar the finish.
 - 3.2. If the pool is empty for any reason, do not allow any walking on the floor, as stains and marks will result.
4. Brush walls and the floor of the pool frequently.
5. Filter(s) requires the pump strainer(s) to be cleaned regularly to eliminate the pump operating without water.
 - 5.1. Leaves become waterlogged, sink to the bottom of the pool, are sucked into the strainer and clog it. This causes the pump to run dry, overheating the motor and damaging the seal.

If you have a sand filter:

1. Only use grade no. 20, or 16/20 filter sand.
2. Periodically check the filter media and remove any accumulation of pine needles, grass cuttings or the like. This type of material may enter the tank through the overdrain distributor and become trapped therein.
3. If an amount of fine sand or similar material has accumulated in the bottom of a new pool prior to start-up, it is recommended that the pool be vacuumed to waste.
4. Pressure sand filter(s) are equipped with an automatic air relief valve to expel all air from the tank interior. Air relief valve should be inspected periodically to ensure proper operations.

If you have a regenerative filter:

1. Only use a premium grade of Perlite or DE for filtration media.
2. Do not use flocculant agents as this can damage the filter elements.

Specific notes for your filter can be found in the filter instruction.

PADDOCK REPORTS

Technical Bulletin 08-01

GREEN BULLETIN

Are your pool operators taking full advantage of having an eco-friendly environment?

It is highly recommended that:

All pools should be vacuumed with a cartridge vacuum regularly to lighten the load on the filter and extend the filter runs.

Pools vacuumed with a cartridge vacuum on a regular basis remove heavy particulates and super fines which may be passing through the filter.

Pools should be brushed at regular intervals at the end of the swimming day to pick up the dirt overnight.

Paddock's Vacuum Sand Filter is a "GREEN" friendly product that is valued at a total of **7 LEED credits** toward their Green project.

LEED Data off of www.arcat.com

● **Water Efficiency**

WE Credit 2 - Innovative wastewater technologies
WE Credit 3 - Water use reduction

● **Energy and Atmosphere**

EA Credit 1 - Optimize energy performance

● **Innovation and Design Process**

ID Credit 1 - Innovative design features

Winterizing

Compak Vacuum Sand Filter

Follow the suggested procedures below:

1. Lower water level in pool below stainless steel gutter.
2. Shut down auto-fill controller and domestic water supply.
3. Once the water level is lowered, close the main drain valve, perimeter overflow valve, return to pool valve to isolate the filter from the pool and open remaining valve(s) to prevent pipes breaking if freeze occurs.
 - a. In high water table areas, we recommend that some water is left in the filter to prevent lifting from hydrostatic pressure from ground water. If this is not an issue please pull water level down to the sand level, turn off pump, refer to step #3, and place a small bilge pump in the backwash trough to remove remaining water.
4. Remove drain plugs from pump box (if available) on filter tank and pump housing.
5. Drain and blow out all auxiliary lines such as chlorinators, heaters, sample lines, etc.
6. Check filter and pipes periodically through winter months for possible excess water in these areas in case of valve leakage or rainwater.
7. Cover pool if possible or remove debris ASAP to prevent surface damage or excess algae.
8. Power down Mark 5, recirculation pump and VFD (if provided). Put the VFD disconnect switch in the off position. Supplemental heat will be required if the room temperature drops below (-20° F).
9. Disconnects for recirculation pump and optional VFD should be lockout / tag out to prevent accidental powering of equipment.

Helpful Hint:

It is recommended that a manual backwash at a full flow rate for an extended time of 5 to 6 minutes is done a minimum of once a year. If your system has the optional air scour feature, it is recommended that the air scour feature is not used during this suggested preventive maintenance backwash. This suggested manual extended flow backwash extends the media life and could prevent having to replace the sand in your filter.

Vacuum Sand Filter

Trouble Shooting

Questions to think about when trouble shooting:

AIR BUBBLES:

1. Are unions on air scour line tight?
2. Are union nuts on ball valve tight?
3. Put second valve on air line after pump.
4. Is ball valve on air scour closed:
 - a. Is air scour valve leaking air?
5. Is MOL level set correctly?
6. Are bolts on pump suction header loose?
7. Is there a leak or restriction in suction line?
8. Is the flange connection on suction box loose?
9. Is the suction box cracked?
10. Is there a cracked flange, pipe, fitting or suction header?
11. Are air scour lines turned in correct position (up)?
12. Inspect air scour pipe and fittings for cracks and leaks.
13. Close #2 pump suction down to 18" vacuum (see if you get air – if so, possible crack in box, ss suction box)
14. Check holes in screen. Add some
15. Check union under equalization screen

PUMP CAVITATES:

1. Is flow rate set? TDH set?
2. Is strainer basket sealed? (not sucking up air)
3. Is there a good seal between pump and strainer?
4. Is filter being back washed properly?
5. Is impeller worn?
6. Is the volute damaged?
7. Are valves set correctly?

Vacuum Sand Filter Trouble shooting – continued:

CLOUDY POOL WATER:

1. Has pool been backwashed properly?
2. Is the filter in filter mode (not by passing) valves 4 & 8 closed securely?
3. Is water chemistry correct? (Alkalinity – PH)?
4. How old is filter media?

REDUCED FLOW RATES:

1. Is UV strainer clean?
2. Has filter been backwashed?
3. Are all valves set correctly? (Any partially closed)?
4. Is air scour being used? (If available).
5. Are pressure gauges reading correct pressure?
6. Is the flow meter operating correctly? (Has it been cleaned)?
7. Is the pump rotation turning correct direction?
8. Are the suction or discharge line restrictions?
9. Is there Impeller blockage or obstruction?
10. Is the Impeller worn?

PUMP LOSES PRIME:

1. Is water level too low? (MOL setting)?
2. Is there an air leak on suction side of pump?
3. Is filter clean? (Has filter been backwashed?)

SAND IN POOL:

1. Was the water level too high when air scour was turned on?
2. Is the filter media correct size and installed with proper freeboard?
3. Does filter have broken laterals?
4. Does filter have a cracked header?

Vacuum Sand Filter Trouble shooting – continued:

HAVE TO BACKWASH TOO OFTEN:

1. Does pool have heavy bather loads?
2. Was back wash performed properly?
3. Have lotions or oils been introduced to pool?
4. Does filter media need changing?

SHUTS DOWN TOO OFTEN FOR BACKWASH:

1. Is vacuum limit switch adjusted to correct setting?
2. Is vacuum limit switch bad?
3. Are valves set properly?
4. Does filter media need changing?

DOES NOT SHUTDOWN FOR BACKWASH:

1. Is vacuum limit switch adjusted to correct setting?
2. Is vacuum limit switch bad?
3. Are controls wired correctly?

PUMP MOTOR RUNNING HOT:

1. Is there low voltage coming to motor?
2. Does pump motor have enough ventilation?
3. Is strainer basket full or restricted?
4. Is the TDH set correctly?
5. Is impeller binding or rubbing?

PUMP MOTOR NOISEY:

1. Does the motor have bad bearings?
2. Is the impeller rubbing?
3. Is strainer basket full or restricted?
4. Is supply or suction restricted? (Check valve settings)?

NO BUBBLES DURING AIR SCOUR:

1. Is valve open to air scour?
2. Is union connected securely?
3. Is there a blockage in the line or fittings?



EQUIPMENT LIST

The following items have been provided by Paddock Pool Equipment Company, Inc.